DSA-110 × CHIME Co-detected FRBs

CHIME-band scattering fits (orientation-corrected)

Nested sampling + Bayesian evidence (M0–M3) · init-independent priors · α fixed = 4.0
16 chan, 0.4–0.8 GHz · outer_trim 0.15, nlive 400 · fit quality judged by residual σ (target = 1)
freq axis was stored descending — every τ here supersedes the earlier (backwards-freq) values

6 clean detections ($\sigma \approx 1$): mahi · oran · johndoeII · whitney · phineas · isha
2 marginal: wilhelm · hamilton 3 fail: chromatica · freya · zach
(freya/zach under-dedispersed — DM, not the fit) casey: refit in progress

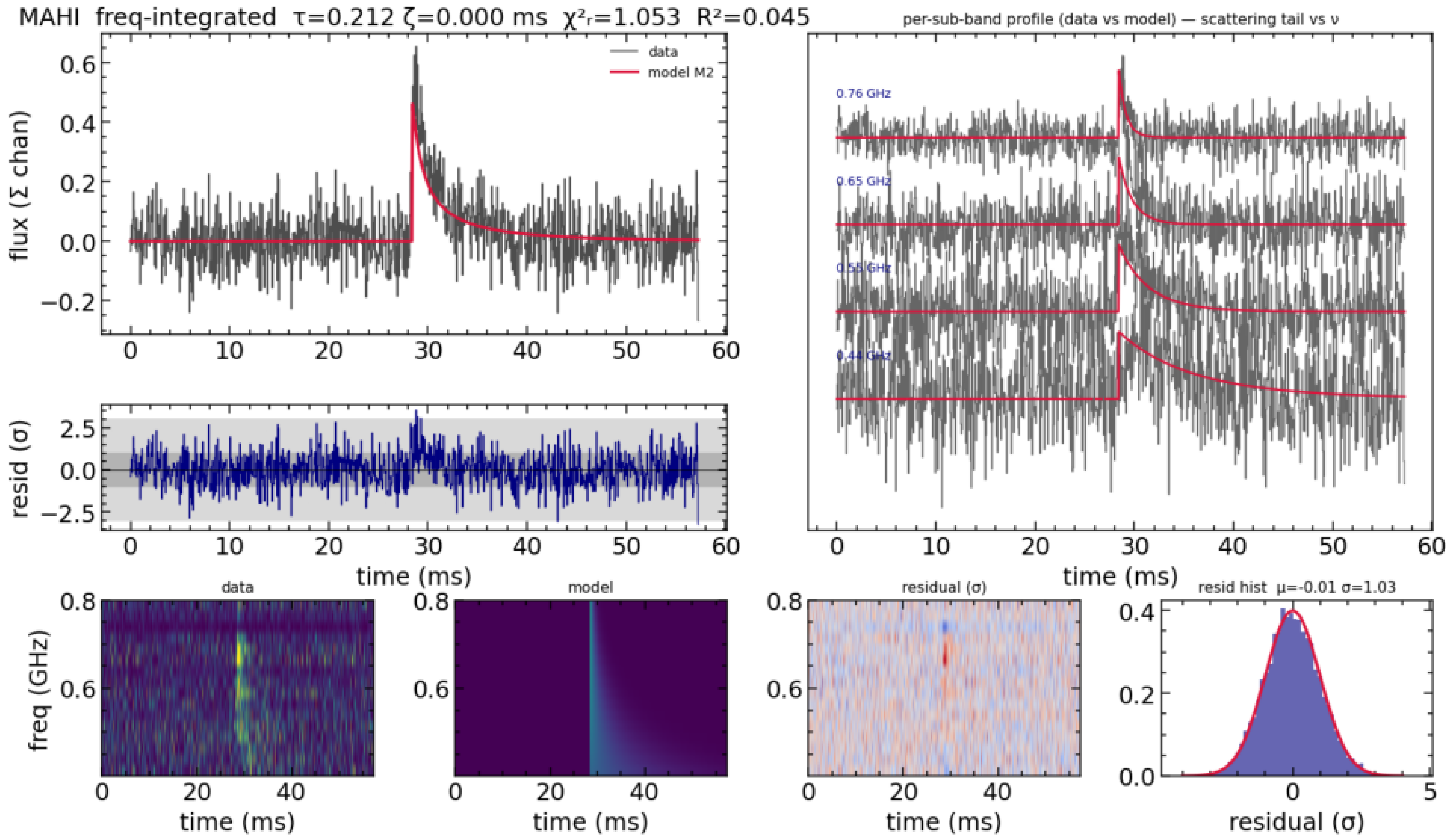
1 slide / burst — fit-quality figure + parameters + verdict

MAHI

best model: M2

detection (M2: scattering-only)

residual σ = 1.03 (target 1.0)



τ_{1GHz} (ms)

0.212 (+0.015/−0.014)

ζ intrinsic (ms)

—

χ^2/dof

1.053

residual σ

1.026

R^2

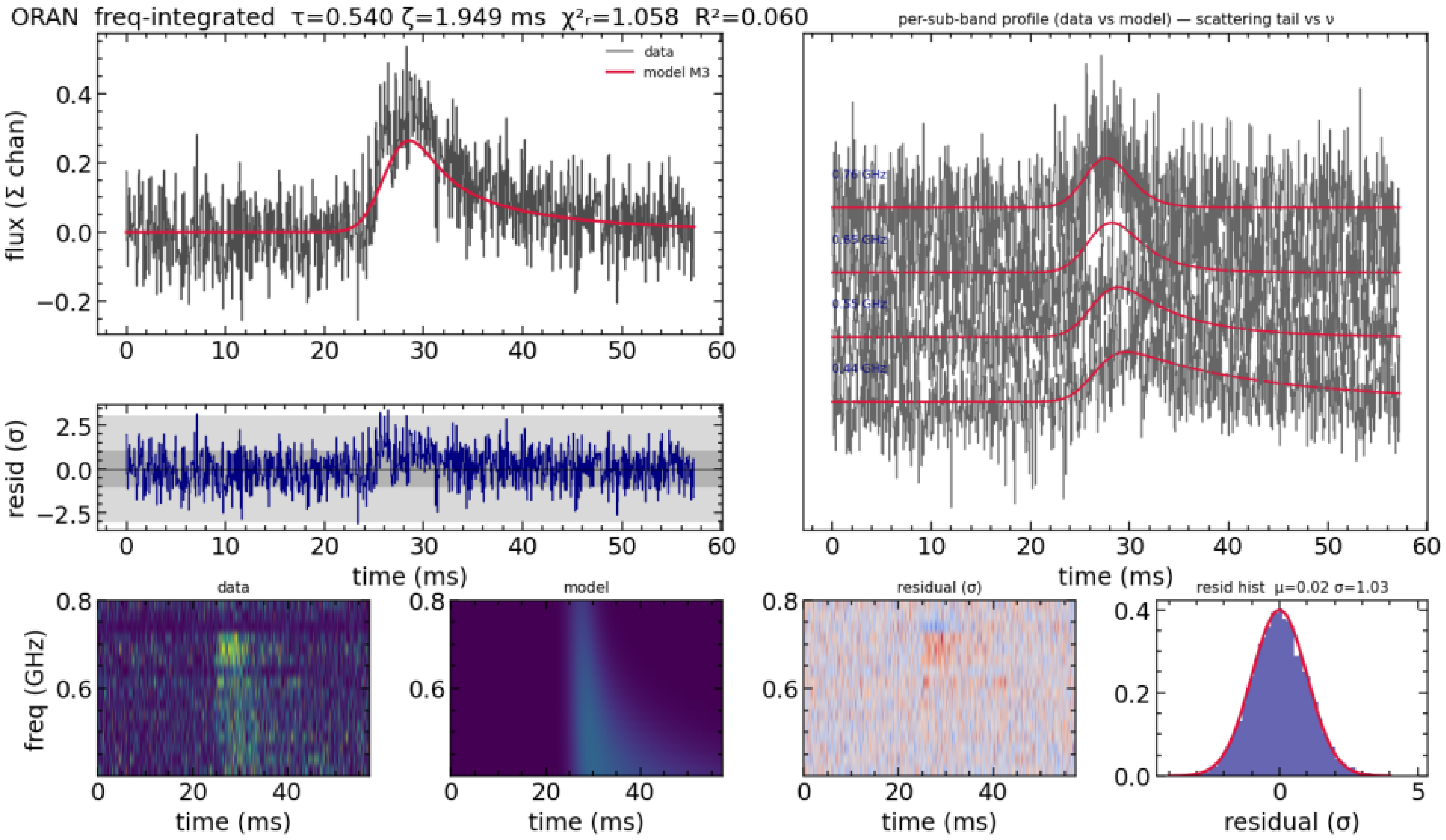
0.045

ORAN

best model: M3

detection

residual σ = 1.03 (target 1.0)



τ_1GHz (ms)

0.54 (+0.063/−0.059)

ζ intrinsic (ms)

1.949 (+0.18/−0.17)

χ^2/dof

1.058

residual σ

1.028

R^2

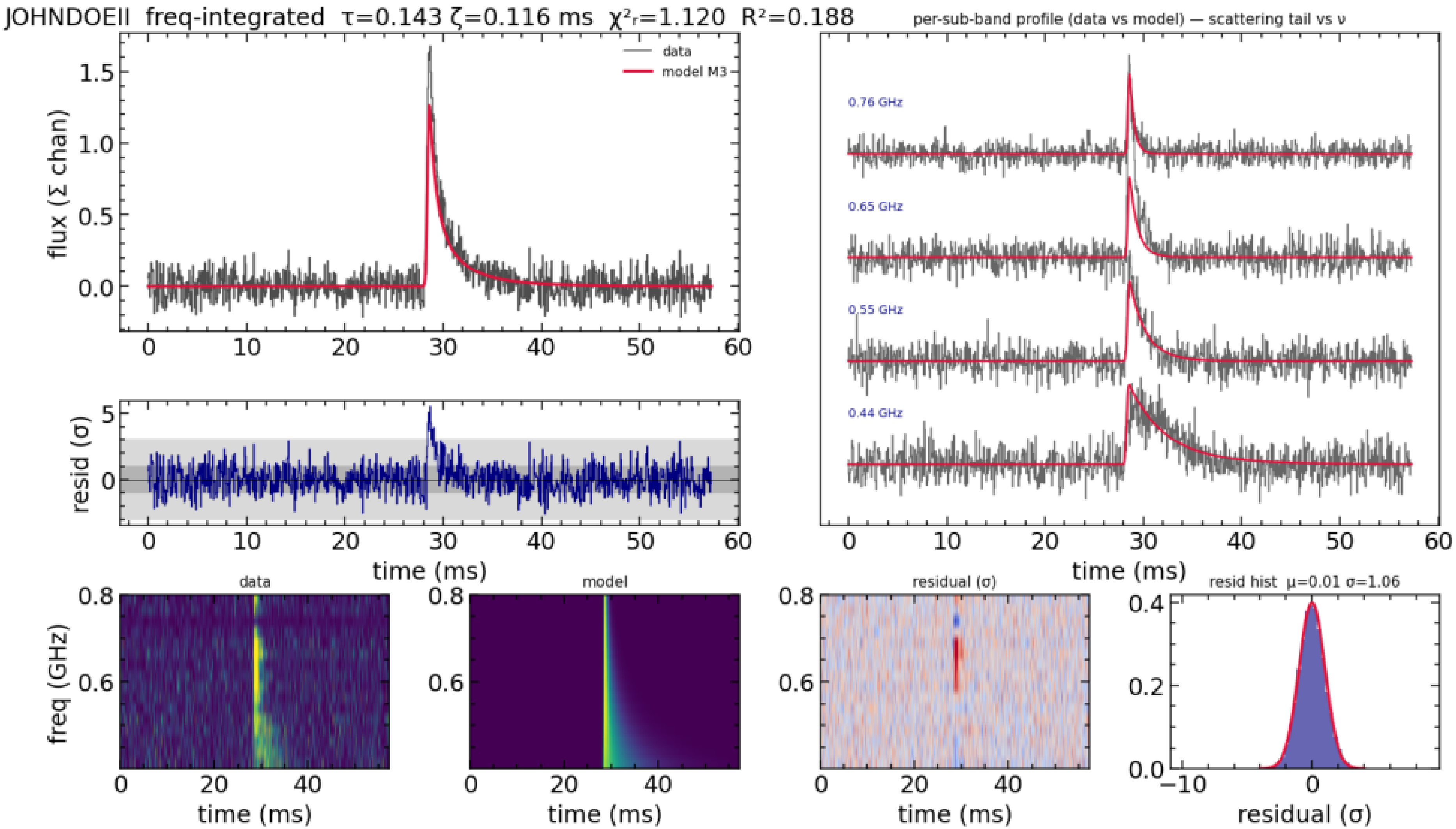
0.060

JOHNDOEII

best model: M3

detection

residual σ = 1.06   (target 1.0)



τ_{1GHz} (ms)

0.143 (+0.0055/−0.0048)

ζ intrinsic (ms)

0.116 (+0.013/−0.012)

χ^2/dof

1.120

residual σ

1.058

R^2

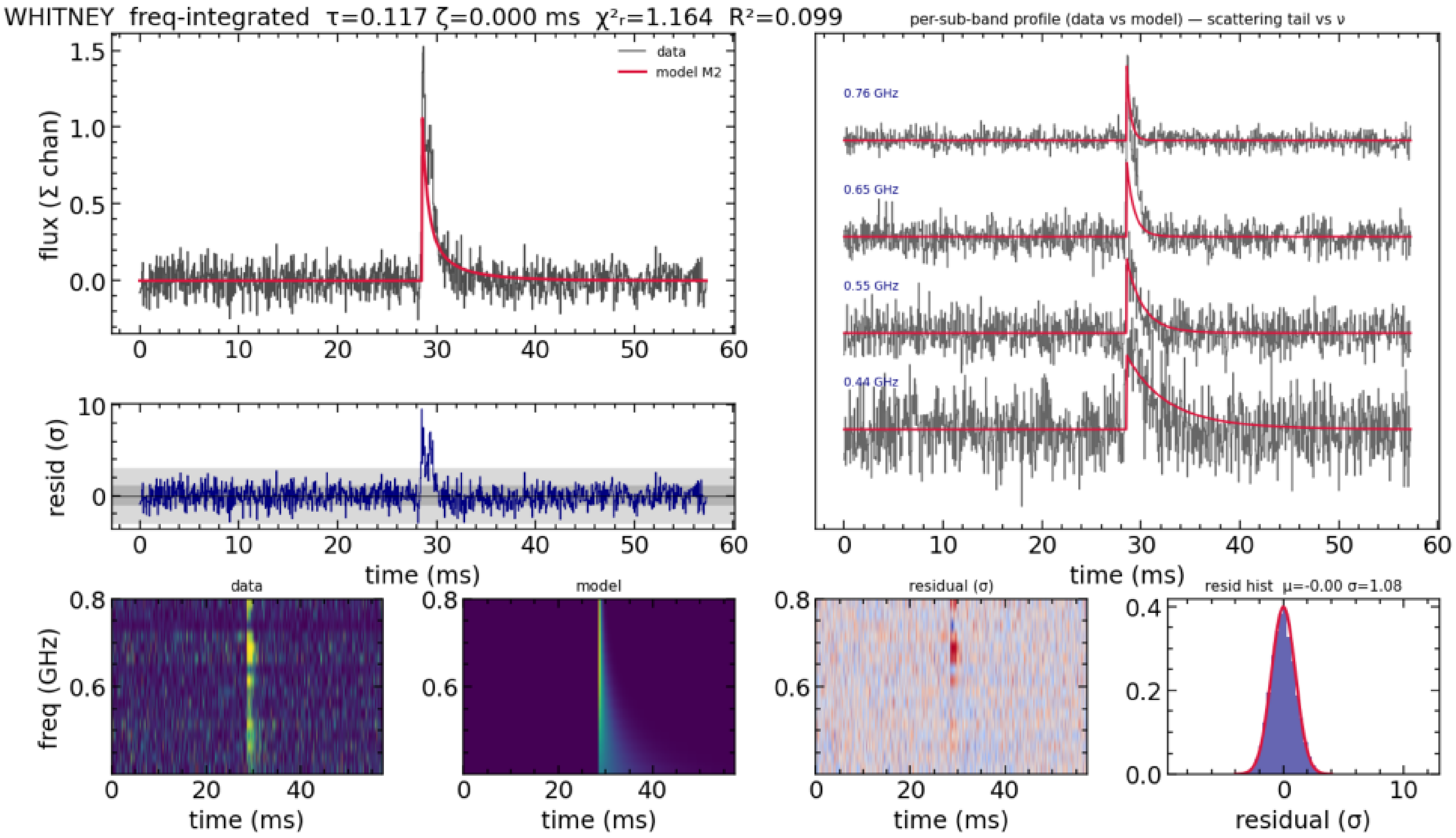
0.188

WHITNEY

best model: M2

detection (M2: scattering-only)

residual σ = 1.08 (target 1.0)



τ₁GHz (ms)

0.117 (+0.0057/−0.0053)

ζ intrinsic (ms)

—

χ^2/dof

1.164

residual σ

1.079

R^2

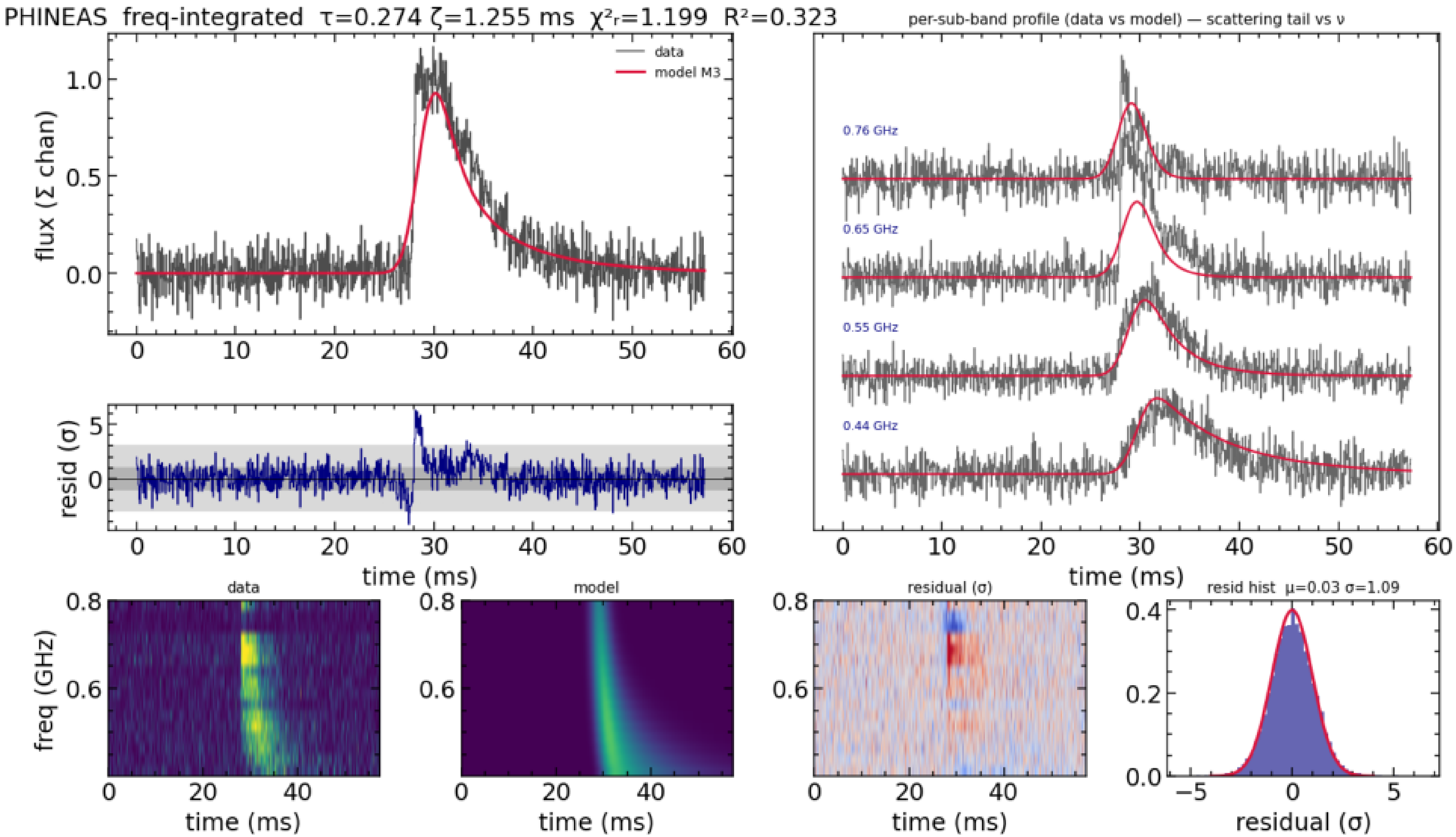
0.099

PHINEAS

detection

best model: M3

residual σ = 1.09   (target 1.0)



τ_{1GHz} (ms)
0.274 (+0.011/−0.011)

ζ intrinsic (ms)
1.255 (+0.045/−0.046)

χ^2/dof
1.199

residual σ
1.095

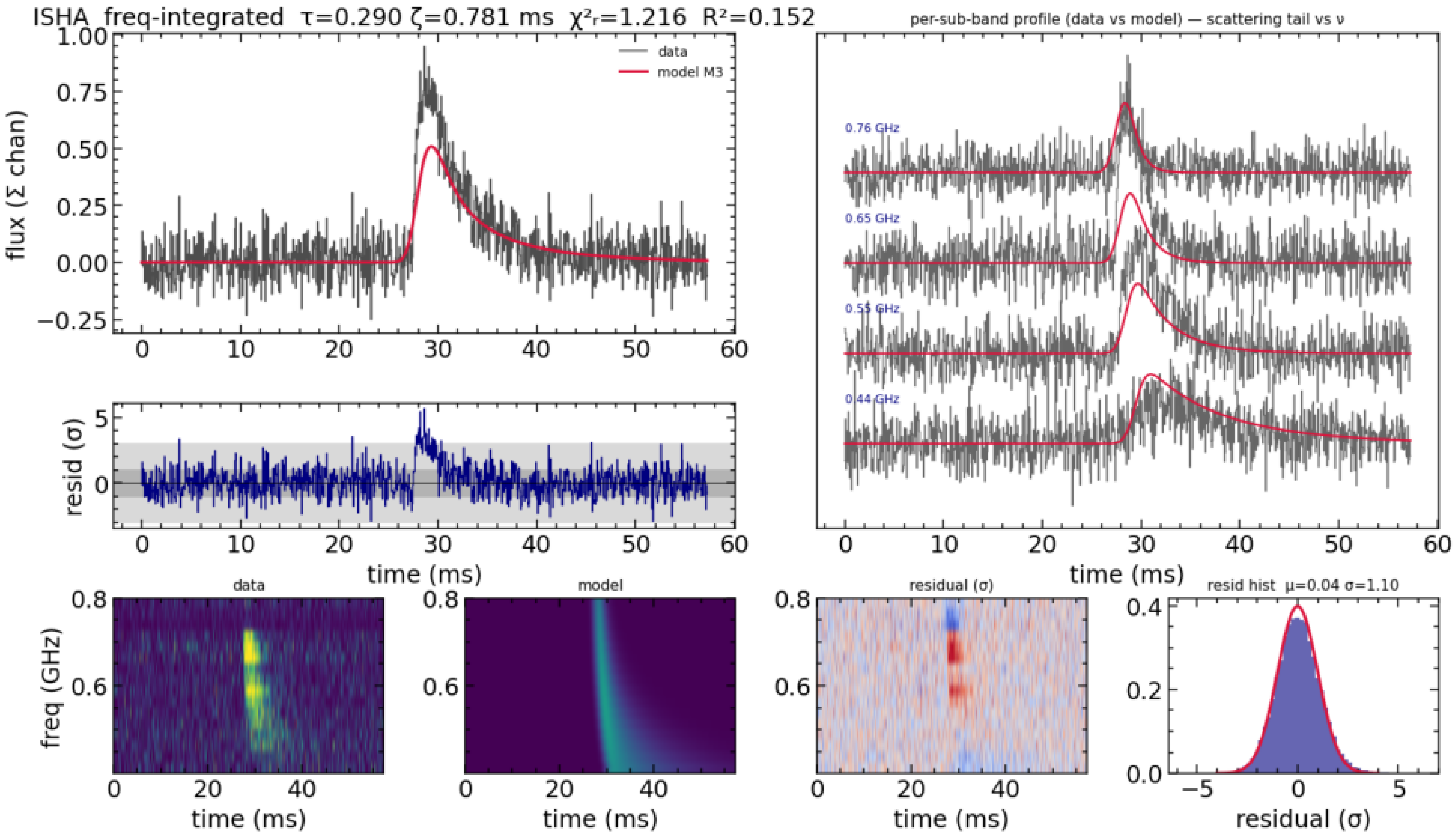
R^2
0.323

ISHA

best model: M3

detection

residual σ = 1.10   (target 1.0)



τ_1GHz (ms)
0.29 (+0.016/−0.014)

ζ intrinsic (ms)
0.781 (+0.047/−0.045)

χ^2/dof
1.216

residual σ
1.102

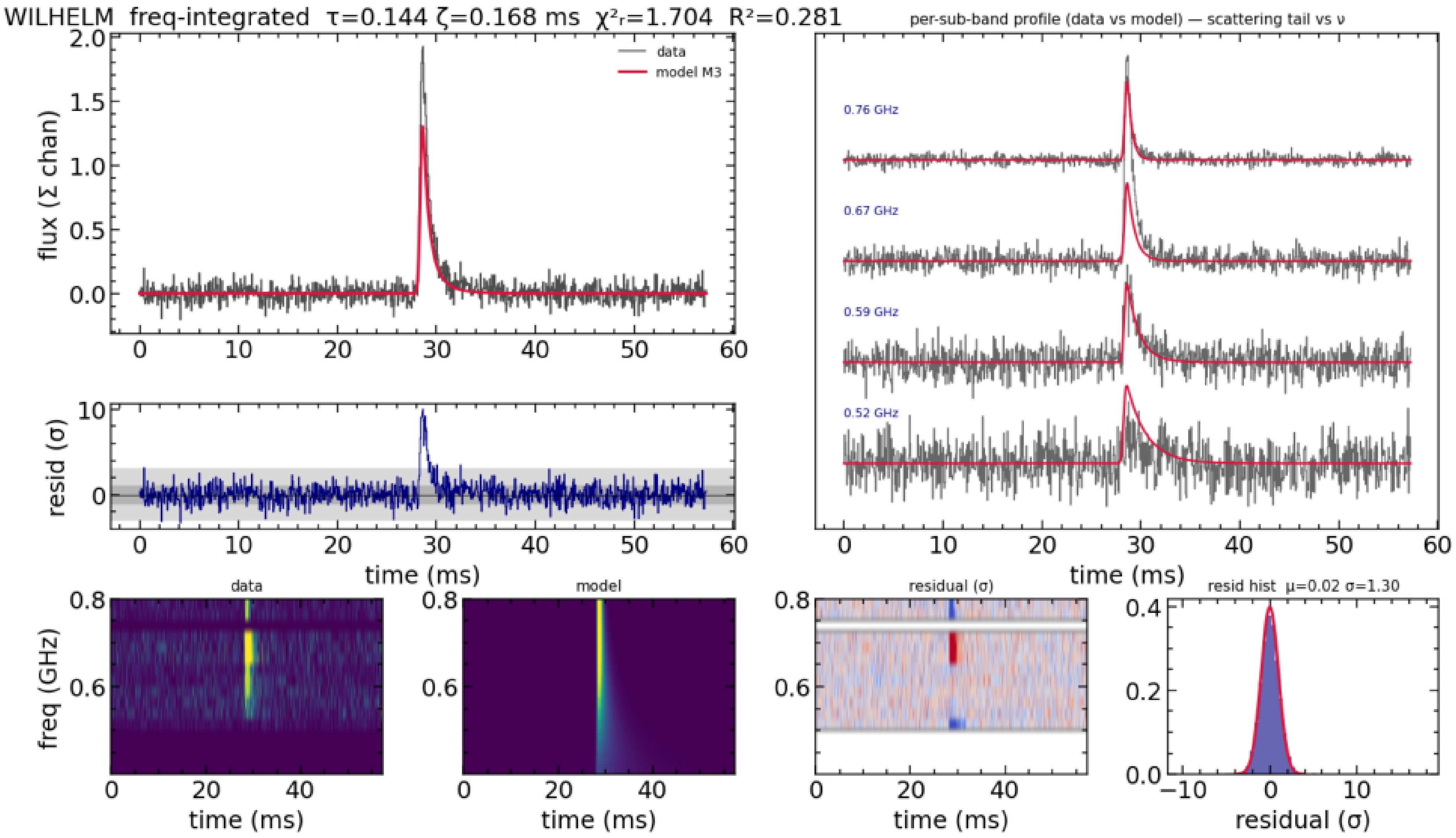
R^2
0.152

WILHELM

marginal

best model: M3

residual σ = 1.30   (target 1.0)



$\tau_{1\mathrm{GHz}}$ (ms)

0.144 (+0.0055/−0.0054)

ζ intrinsic (ms)

0.168 (+0.009/−0.009)

χ^2/dof

1.704

residual σ

1.305

R^2

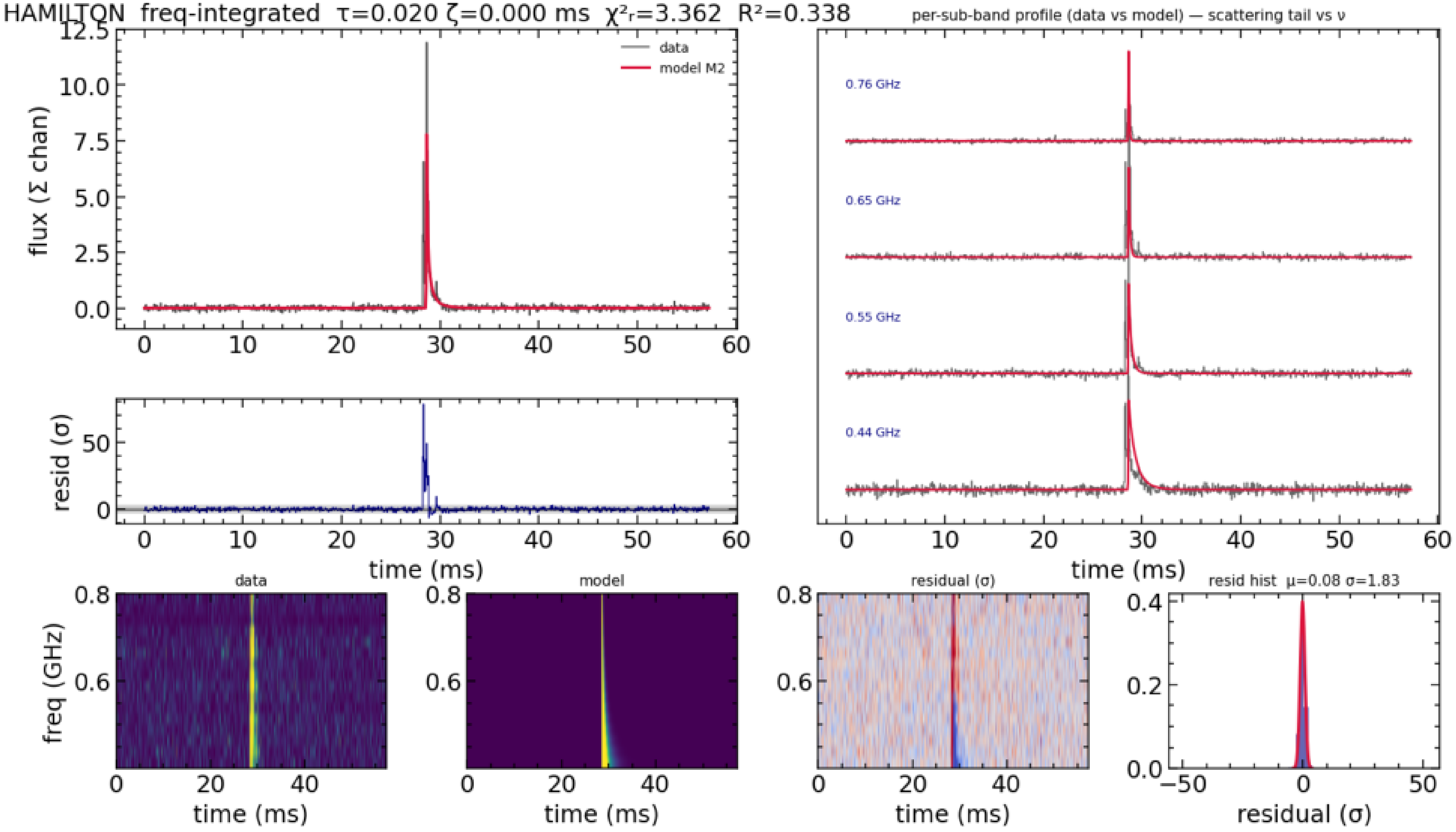
0.281

HAMILTON

marginal (M2)

best model: M2

residual σ = 1.83 (target 1.0)



τ_{1GHz} (ms)

0.02 (+0.0002/−0.0002)

ζ intrinsic (ms)

—

χ^2/dof

3.362

residual σ

1.831

R^2

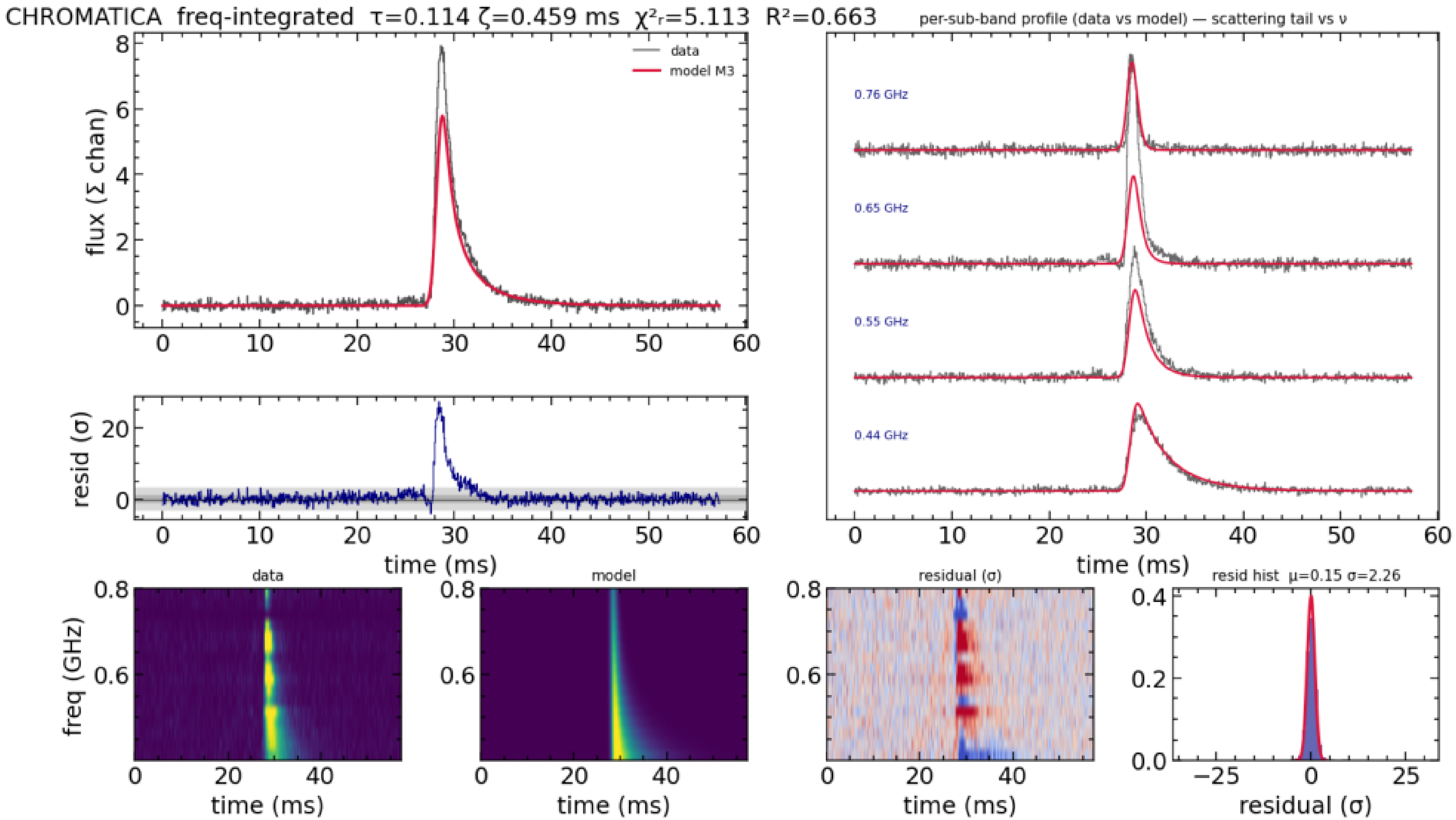
0.338

CHROMATICA

best model: M3

poor fit

residual σ = 2.26   (target 1.0)



τ_{1GHz} (ms)

0.114 (+0.0009/−0.0008)

ζ intrinsic (ms)

0.459 (+0.004/−0.004)

χ^2/dof

5.113

residual σ

2.256

R^2

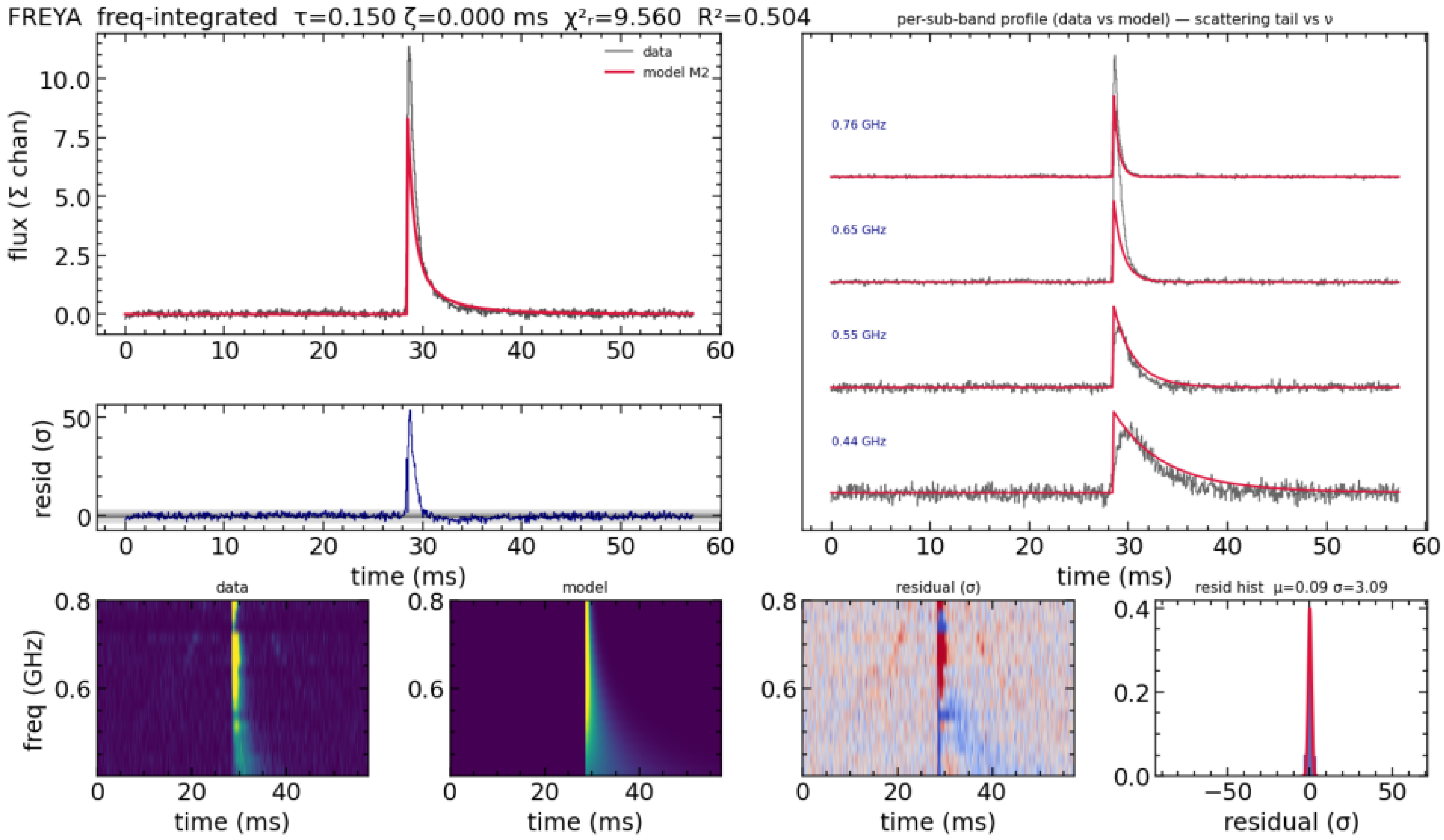
0.663

FREYA

best model: M2

FAIL — under-dedispersed (DM sweep)

residual σ = 3.09   (target 1.0)



τ₁GHz (ms)

0.15 (+0.0006/−0.0007)

ζ intrinsic (ms)

—

χ^2/dof

9.560

residual σ

3.090

R^2

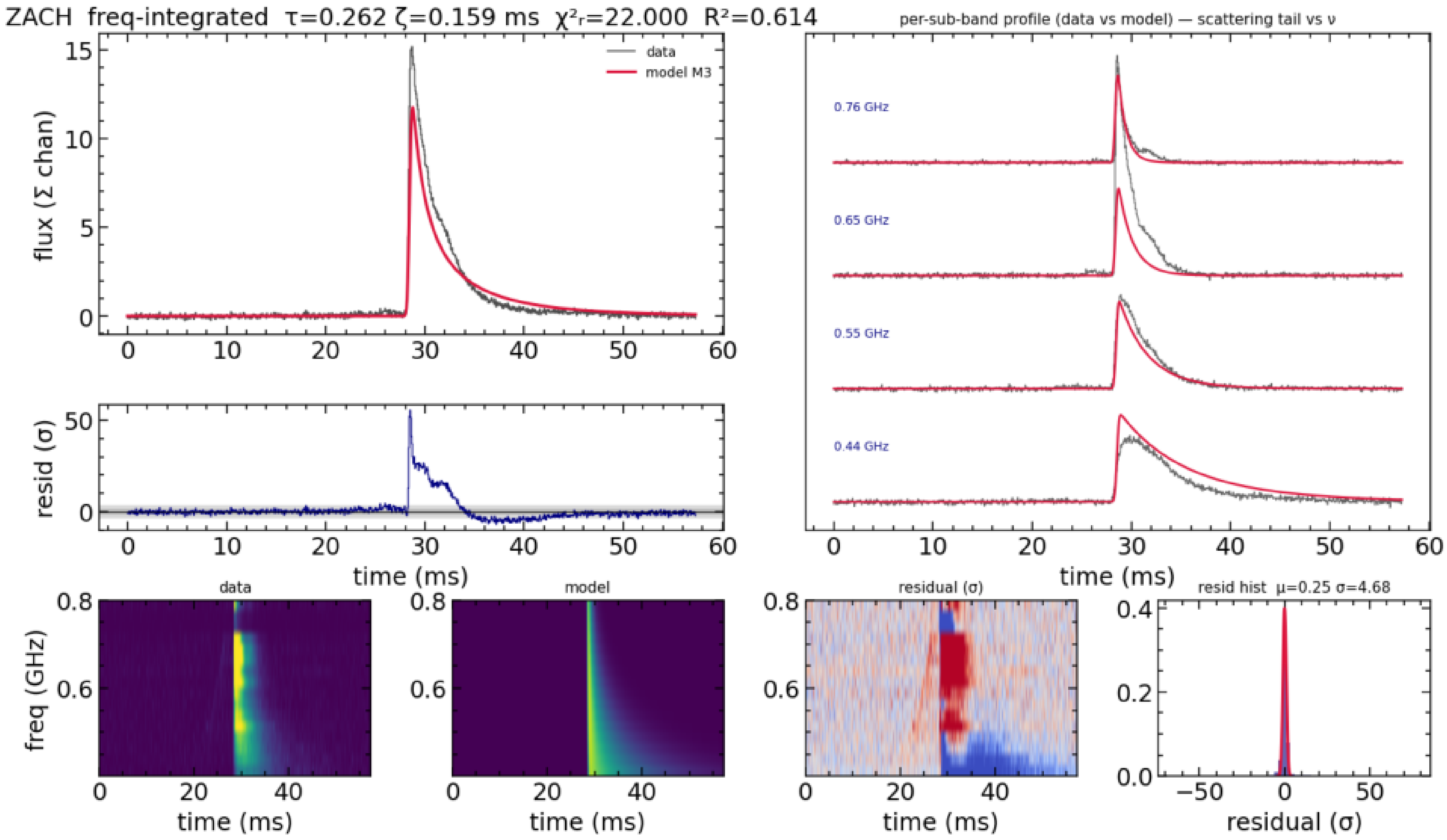
0.504

ZACH

best model: M3

FAIL — under-dedispersed (DM sweep)

residual σ = 4.68   (target 1.0)



τ_{1GHz} (ms)

0.262 (+0.0009/−0.0009)

ζ intrinsic (ms)

0.159 (+0.002/−0.002)

χ^2/dof

22.000

residual σ

4.683

R^2

0.614

CASEY

refit re-running orientation-corrected (M3 stalls at noise floor; M1 strongly preferred). Slide updates when it lands.